

Project Design Phase - Proposed Solution Template

Date	Team ID	Project Name	Maximum Marks
18 June 2026		ShopEZ E-Commerce Platform	2 Marks

1. Proposed Solution Matrix

Parameter	Description
Problem Statement	Customers encounter friction due to slow cart loading, unclear order billing (taxes, delivery charges), and administrative delays in updating store catalogs.
Idea / Solution Description	ShopEZ is a responsive full-stack MERN application. It uses custom React Hooks/Context to manage cart items instantly in localStorage, displays exact GST and shipping costs, and provides an admin control panel to perform CRUD operations on products.
Novelty / Uniqueness	Zero page-reloads during catalog browsing and sorting, dynamic price-range filters, and instant local token invalidation upon API authentication failure (401 response).
Social Impact / Customer Satisfaction	Low bandwidth accessibility, support for Cash on Delivery (COD) for users without banking access, and complete transaction transparency.
Business Model (Revenue Model)	Platform commissions on product sales, direct sales from local inventories, and premium delivery subscription options.
Scalability of the Solution	Deployable as isolated client (Vercel/Netlify) and backend services (Render/AWS ECS/Lambda) with MongoDB Atlas providing multi-region scaling.